Spatial Machine Learning in R

Integration of spatial data of different types and granularity

Session 2

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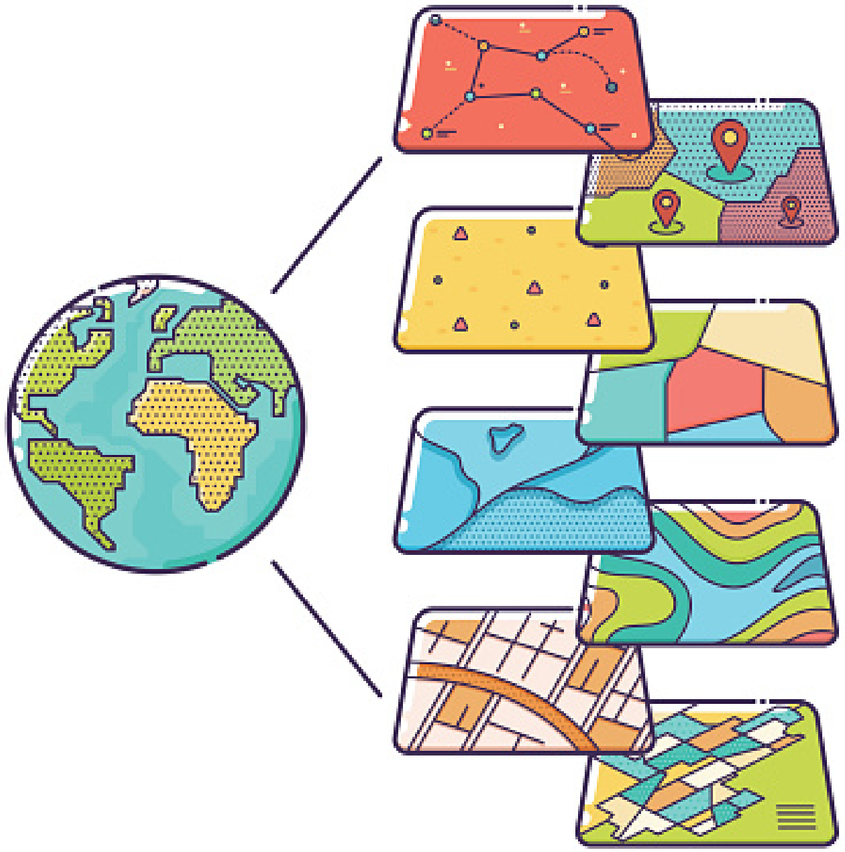
Why do we need spatial data integration?

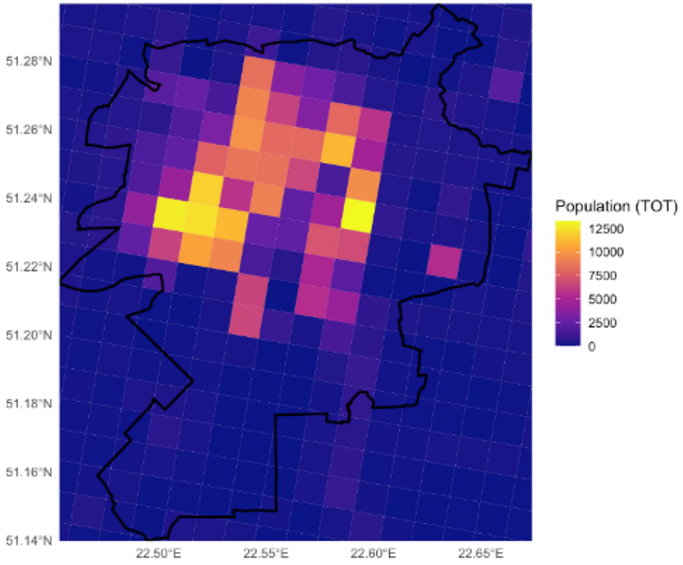
- Real-world data comes in different forms
- Different spatial scales
- Different geometry types
- To answer spatial questions, we must combine them



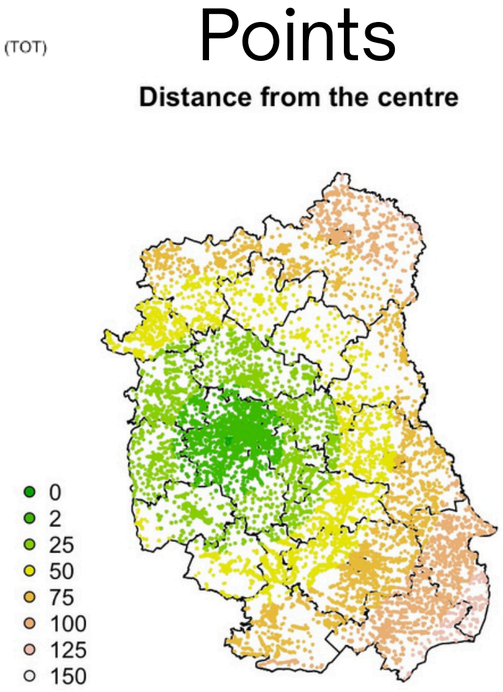
Source: https://www.istockphoto.com/pl/wektor/gis-koncepcja-warstwy-danych-dla-grafika-informacyjna-gm626627638-110754217

Some of the spatial data types

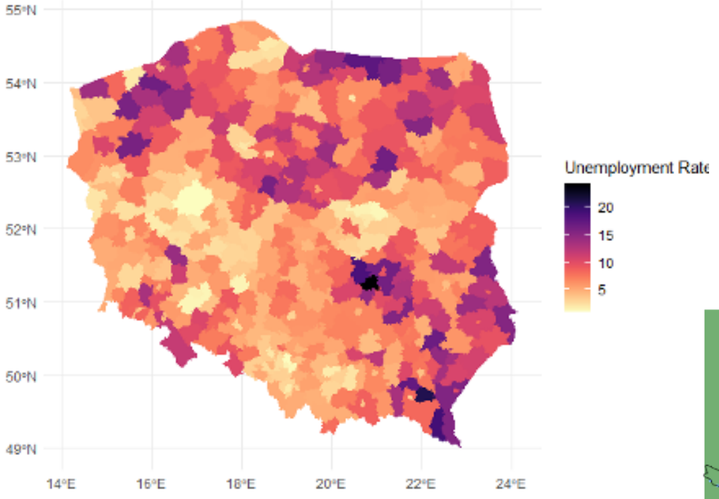
Grids



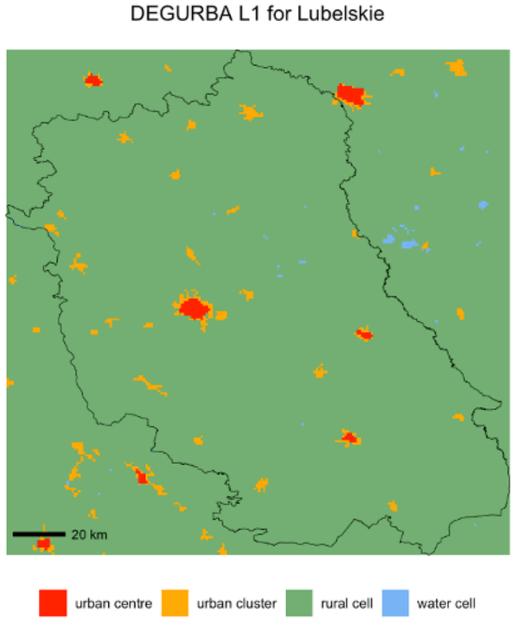
Points



Polygons



Rasters

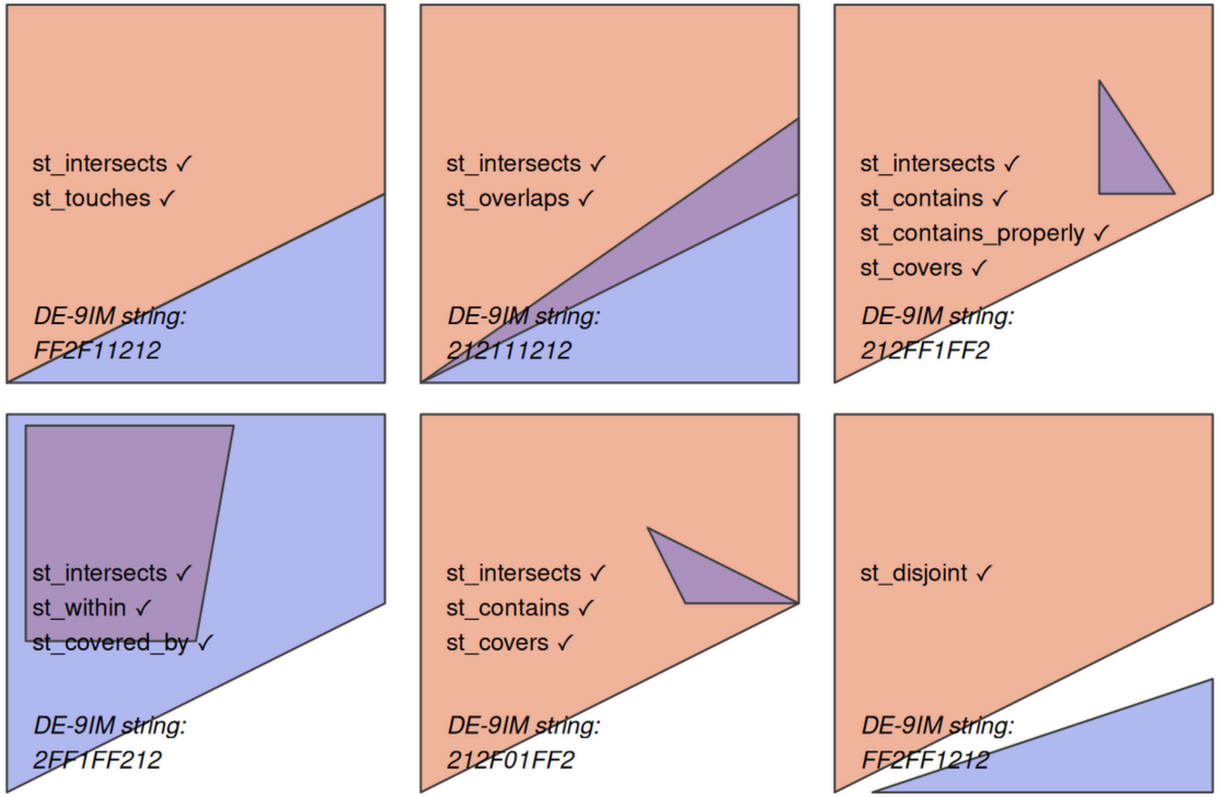


What counts as a spatial match?

Your analytical result depends on this choice!

Common spatial predicates:

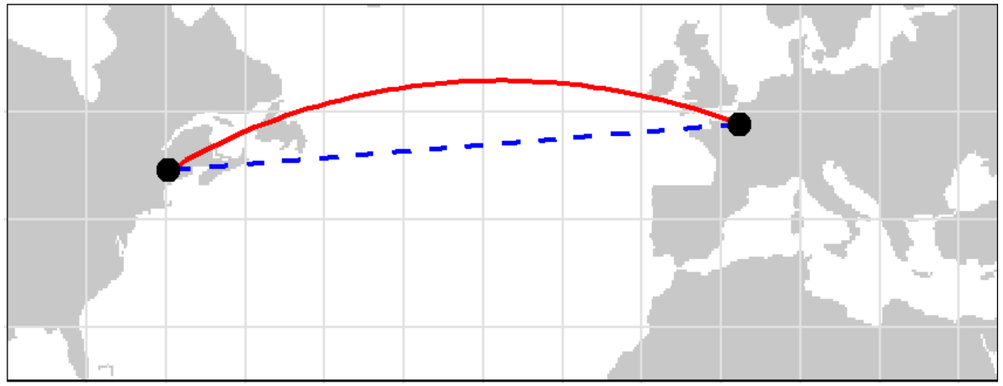
- **st_within**(A, B) → A lies strictly inside B (border excluded)
- **st_intersects**(A, B) → A and B share any space (including borders)
- **st_contains**(A, B) → B lies inside A (inverse of within)
- **st_overlaps**(A, B) → A and B partially overlap (but neither fully contains the other)
- **All options**: https://r-spatial.github.io/sf/reference/geos_binary_pred.html



Source: https://r.geocompx.org/spatial-operations

Coordinate systems & units

Before any spatial operation we must make sure that **all layers must use the same CRS!** Otherwise spatial joins, overlays, and distances may fail or give wrong results.



Source: https://pygis.io/docs/d_crs_what_is_it.html

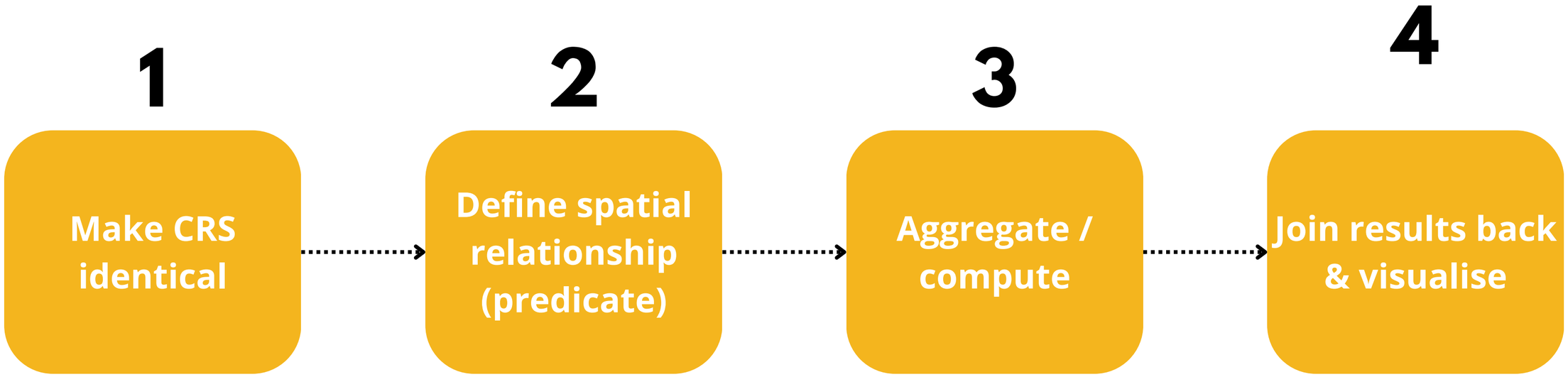
Geographic CRS (e.g. EPSG:4326)

- Coordinates in **degrees**
- Based on latitude and longitude
- Good for global positioning
- Not suitable for area or distance calculations
- 1 degree ≠ constant distance on Earth

Projected CRS (e.g. EPSG:2180)

- Coordinates in **meters** (or feet)
- Flat map projection
- Suitable for: distance calculations, area calculations etc
- Units are consistent and interpretable

The core workflow pattern

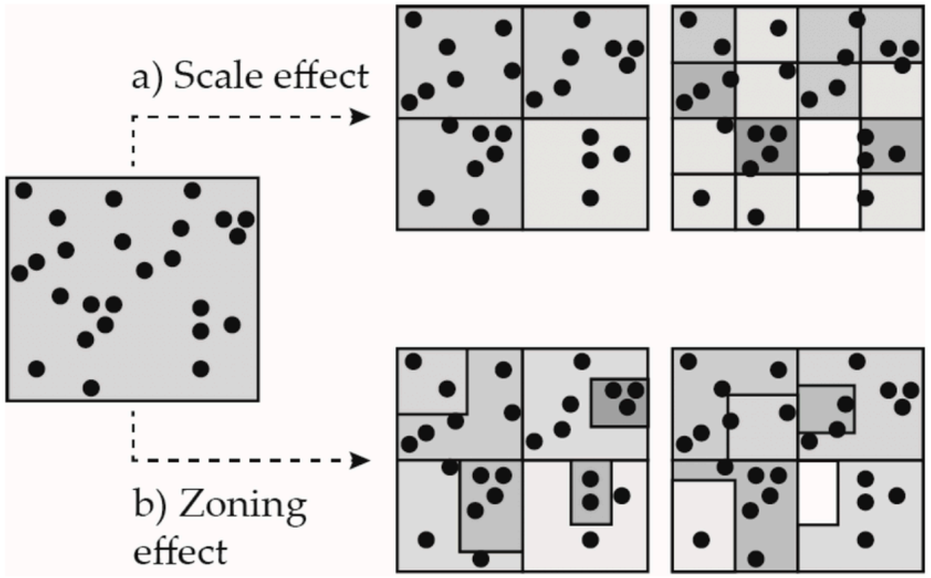


Scale changes results (MAUP concept)

MAUP (Modifiable Areal Unit Problem) refers to the fact that statistical results can change depending on how spatial data is aggregated.

Two Components of MAUP

- **Scale Effect -** results change when we use different levels of aggregation (e.g. grid cells vs poviats vs voivodeships)
- **Zone Effect -** results change when we redraw boundaries at the same scale (different grouping of the same basic units)



Source: https://gdsl-ul.github.io/san/02-spatial_data.html